CBSE Class 12 Pe — Half-Yearly Predicted Paper (Set 4 of 10)

SECTION A: Objective Type Questions (18 Marks: 18 x 1)

All questions are compulsory. Consists of MCQs and Assertion-Reason questions.

Q1. In tournament fixtures, the procedure of placing highly ranked or strong teams directly into the quarter-finals or semi-finals so they do not get eliminated early is known as: (A) Bye (B) Seeding (C) Special Seeding (D) Wild Card Entry *Unit 1: Management of Sporting Events | [PYQ] | Tag: Very High***

Q2. Which postural deformity is characterized by an abnormal outward curvature of the legs, resulting in a widened gap between the knees when standing with feet together? (A) Knock Knees (Genu Valgum) (B) Bow Legs (Genu Varum) (C) Flat Foot (D) Lordosis *Unit 2: Children & Women in Sports | [PYQ] | Tag: Very High***

Q3. Which of the following asanas is famously translated as the "Half Lord of the Fishes Pose" and is highly effective in managing Diabetes? (A) Ardha Matsyendrasana (B) Paschimottanasana (C) Bhujangasana (D) Gomukhasana *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | [PYQ] | Tag: High***

Q4. The headquarters of the International Paralympic Committee (IPC), the global governing body of the Paralympic Movement, is located in: (A) Lausanne, Switzerland (B) London, United Kingdom (C) Athens, Greece (D) Bonn, Germany *Unit 4: Physical Education & Sports for CWSN | [Practice] | Tag: Medium***

Q5. Which of the following is an essential Micro-nutrient required by the human body for proper thyroid function? (A) Carbohydrate (B) Protein (C) Iodine (D) Fat *Unit 5: Sports & Nutrition | [PYQ] | Tag: Very High***

Q6. The 50m Dash (Standing Start) in the SAI Khelo India Fitness Test battery is primarily used to measure an athlete's: (A) Cardiovascular Endurance (B) Explosive speed and acceleration (C) Lower body flexibility (D) Upper body strength *Unit 6: Test and Measurement in Sports | [PYQ] | Tag: Very High***

Q7. A joint injury where the articulating bones are violently forced out of their normal anatomical position is medically termed as: (A) Strain (B) Sprain (C) Dislocation (D) Contusion *Unit 7: Physiology & Injuries in Sport | [PYQ] | Tag: Very High***

Q8. The physical force of friction always acts in the __ direction of the moving object. (A) Upward (B) Downward (C) Same (D) Opposite *Unit 8: Biomechanics and Sports | [Practice] | Tag: Very High***

Q9. Competitions held strictly outside the boundaries of an institution, involving teams from various different schools or colleges, are known as: (A) Intramural tournaments (B)

Extramural tournaments (C) House matches (D) Domestic events *Unit 1: Management of Sporting Events | [PYQ] | Tag: Very High***

Q10. The core components of the "Female Athlete Triad" include Eating Disorders, Amenorrhea, and: (A) Dysmenorrhea (B) Osteoporosis (C) Lordosis (D) Spondylitis *Unit 2: Children & Women in Sports | [PYQ] | Tag: Very High***

Q11. Severe deficiency of Vitamin A in the human diet primarily leads to which of the following conditions? (A) Scurvy (B) Night Blindness (C) Rickets (D) Beriberi *Unit 5: Sports & Nutrition | [Practice] | Tag: Very High***

Q12. What does the abbreviation ASD stand for in the context of cognitive disabilities? (A) Automatic Sensory Disorder (B) Autism Spectrum Disorder (C) Attention Sensory Deficit (D) Auditory Spectrum Disability *Unit 4: Physical Education & Sports for CWSN | [PYQ] | Tag: Very High***

Q13. Which of the following non-nutritive components of a diet provides a sweet taste but contains zero caloric energy, often used for weight control? (A) Roughage (B) Water (C) Artificial Sweeteners (D) Plant compounds (Phytochemicals) *Unit 5: Sports & Nutrition | [PYQ] | Tag: High***

Q14. In the Rikli & Jones Senior Citizen Fitness Test, which specific test evaluates the upper body flexibility of elderly individuals? (A) Chair Sit and Reach Test (B) Back Scratch Test (C) Arm Curl Test (D) 8-Foot Up and Go Test *Unit 6: Test and Measurement in Sports | [PYQ] | Tag: Very High***

Q15. The normal stroke volume for a healthy, untrained adult at rest is approximately: (A) 50 - 70 ml/beat (B) 90 - 110 ml/beat (C) 120 - 140 ml/beat (D) 150 - 170 ml/beat *Unit 7: Physiology & Injuries in Sport | [Practice] | Tag: High***

Q16. Assertion (A): In a knock-out tournament, "Byes" are given in the first round to ensure that the number of teams playing in subsequent rounds becomes a perfect power of two. **Reason (R):** Giving byes prevents uneven match pairings and ensures a symmetrical fixture bracket up to the finals. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 1: Management of Sporting Events | [Practice] | Tag: Very High***

Q17. Assertion (A): Water is considered an essential component of an athlete's diet despite providing zero caloric energy. **Reason (R):** Water regulates body temperature through sweating and transports vital nutrients and oxygen to the working muscles. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 5: Sports & Nutrition | [PYQ] | Tag: Very High***

Q18. Assertion (A): Increasing an athlete's base of support significantly reduces their physical stability on the field. **Reason (R):** The broader the base of support, the harder it is for the line of gravity to fall outside of it, thus improving equilibrium. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 8: Biomechanics and Sports | [PYQ] | Tag: High***

SECTION B: Very Short Answer Type (10 Marks: Attempt any 5 out of 6 x 2)

Word limit: 60-90 words.

Q19. Briefly explain the concept of 'Seeding' in tournament fixtures. How does it help maintain the competitive quality of a tournament? *Unit 1: Management of Sporting Events | [PYQ] | Tag: Very High***

Q20. State any two major physiological or environmental causes that lead to the postural deformity known as 'Lordosis'. *Unit 2: Children & Women in Sports | [PYQ] | Tag: High***

Q21. Mention any two health benefits of practicing *Trikonasana* (Triangle Pose). *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | [PYQ] | Tag: Very High***

Q22. What is the primary role of an Occupational Therapist in assisting Children with Special Needs (CWSN)? *Unit 4: Physical Education & Sports for CWSN | [Practice] | Tag: High***

Q23. Define 'Laceration'. Give one common example of how this soft tissue injury can occur in sports. *Unit 7: Physiology & Injuries in Sport | [PYQ] | Tag: Very High***

Q24. Differentiate clearly between Static Equilibrium and Dynamic Equilibrium with suitable examples from sports. *Unit 8: Biomechanics and Sports | [Practice] | Tag: Very High***

SECTION C: Short Answer Type (15 Marks: Attempt any 5 out of 6 x 3)

Word limit: 100-150 words.

Q25. Write the mathematical formulas to calculate the total number of matches and total rounds in a Single League tournament. Using these formulas, calculate the exact number of matches and rounds if 8 teams participate. *Unit 1: Management of Sporting Events | [PYQ] | Tag: Very High***

Q26. Explain the concept of the 'Female Athlete Triad'. Briefly discuss 'Eating Disorders' as a core component of this syndrome. *Unit 2: Children & Women in Sports | [PYQ] | Tag: Very High***

Q27. Enlist and explain any three major 'Food Myths' regarding nutrition and weight loss that are prevalent among athletes and the general public. *Unit 5: Sports & Nutrition | [PYQ] | Tag: High***

Q28. Briefly explain the administration procedure of the 600m Run/Walk test. What specific component of physical fitness does it measure? *Unit 6: Test and Measurement in Sports | [PYQ] | Tag: Very High***

Q29. Discuss any three immediate (short-term) physiological effects of rigorous exercise on the human cardiovascular system. *Unit 7: Physiology & Injuries in Sport | [PYQ] | Tag: High***

Q30. State Newton's First Law of Motion (Law of Inertia). Explain its practical application by providing two distinct examples from the field of sports. *Unit 8: Biomechanics and Sports | [PYQ 2024] | Tag: Very High***

SECTION D: Case-Study Based Questions (12 Marks: 3 x 4)

All 3 questions are compulsory. Internal choice is provided in one sub-part of each.

Q31. The Sports Authority of India (SAI) recommends the Khelo India Fitness Test battery for primary school children (Age group 5-8 years / Classes 1 to 3) to evaluate their fundamental motor skills and body composition early on. A primary school has set up various testing stations on its sports ground. **(i)** Which specific test from this battery should the instructors use to measure the coordination and speed of the upper limbs? **(ii)** Name the specific test conducted to assess a child's static balance. **(iii)** What is the mathematical formula used to calculate a child's Body Mass Index (BMI) at the BMI station? **(iv)** Why is assessing physical fitness crucial at such a young age? **OR** State the duration or scoring method for the Flamingo Balance Test. *Unit 6: Test and Measurement in Sports | [PYQ] | Tag: Very High***

Q32. During the final lap of a 400m sprint event, an athlete experiences a severe burning sensation in his leg muscles and starts breathing heavily. Desperate to cross the finish line, he pushes past his limit but suddenly collapses holding his hamstring, suffering a painful muscle tear. **(i)** The burning sensation in the working muscles during an intense sprint is primarily due to the rapid accumulation of _. ***(ii)*** *A severe tear or rupture in a muscle or tendon (like the hamstring) is medically classified as a _.* **(iii)** The heavy breathing during the sprint naturally increases the volume of air inhaled and exhaled per breath. What is this volume called? **(iv)** What immediate first-aid management protocol should the coach apply to the athlete's torn hamstring? **OR** The amount of blood pumped by the heart in one minute is known as? *Unit 7: Physiology & Injuries in Sport | [PYQ] | Tag: Very High***

Q33. To foster an inclusive environment, a reputed school organizes an adaptive Sports Day for its mainstream students and Children with Special Needs (CWSN). The physical education department provides auditory balls with internal bells for visually impaired children and uses visual flags instead of starting whistles for hearing-impaired children. **(i)** Providing an auditory ball with bells is an example of modifying the _ *to make sports accessible.* ***(ii)*** *Which international organization promotes and governs sporting events exclusively for deaf and hearing-impaired athletes?* ***(iii)*** *The educational philosophy of teaching disabled children alongside their mainstream peers in the same environment is called _.* **(iv)** Name the specific rehabilitation professional who assists CWSN in improving fine motor skills (e.g., proper grip to hold a ball). **OR** What is the minimum age requirement for an athlete to participate in the Special Olympics? *Unit 4: Physical Education & Sports for CWSN | [PYQ] | Tag: High***

SECTION E: Long Answer Type (15 Marks: Attempt any 3 out of 4 x 5)

Word limit: 200-300 words.

Q34. Draw a comprehensive Knock-out fixture for 23 teams participating in a district basketball tournament. Clearly display all your mathematical calculations including the number of matches, number of rounds, total byes, and the specific distribution of byes in the upper and lower halves. *Unit 1: Management of Sporting Events | [PYQ] | Tag: Very High***

Q35. What do you understand by 'Diabetes'? Enlist any four Yoga Asanas recommended in the syllabus for its management. Thoroughly describe the step-by-step procedure and two contraindications of *Paschimottanasana*. *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | [PYQ] | Tag: Very High***

Q36. Define 'Vitamins'. Classify vitamins into Fat-Soluble and Water-Soluble categories, listing the specific alphabetical names of vitamins under each. Furthermore, state at least one disease caused by the severe deficiency of Vitamin A, Vitamin C, and Vitamin D. *Unit 5: Sports & Nutrition | [PYQ] | Tag: High***

Q37. What are 'Levers' in biomechanics? Thoroughly explain the three classes of levers (Class I, II, and III). Provide the structural definition (positions of Fulcrum, Effort, and Load) and one anatomical or sports example for each class. *Unit 8: Biomechanics and Sports | [PYQ 2024] | Tag: Very High***

Answer Key

Q1. (C) Special Seeding **Q2.** (B) Bow Legs (Genu Varum) **Q3.** (A) Ardha Matsyendrasana **Q4.** (D) Bonn, Germany **Q5.** (C) Iodine **Q6.** (B) Explosive speed and acceleration **Q7.** (C) Dislocation **Q8.** (D) Opposite **Q9.** (B) Extramural tournaments **Q10.** (B) Osteoporosis **Q11.** (B) Night Blindness **Q12.** (B) Autism Spectrum Disorder **Q13.** (C) Artificial Sweeteners **Q14.** (B) Back Scratch Test **Q15.** (A) 50 - 70 ml/beat **Q16.** (A) Both (A) and (R) are true and (R) is the correct explanation of (A). **Q17.** (A) Both (A) and (R) are true and (R) is the correct explanation of (A). **Q18.** (D) (A) is false, but (R) is true. (Reason: A wider base *increases* stability; center of gravity must fall *inside* the base of support for equilibrium).

Q19. Seeding is a procedure in tournament draws where top-ranking or highly skilled teams are strategically placed in different halves or quarters of the fixture. This prevents strong teams from playing and eliminating each other in the initial rounds, ensuring the later stages (semi-finals, finals) remain highly competitive. **Q20.** 1. Congenital spinal defects or weak lower back muscles. 2. Obesity and carrying excessive weight around the abdomen, which pulls the lower spine forward into an exaggerated inward curve. **Q21.** 1. It burns fat and reduces waistline obesity. 2. It stretches and strengthens the spine, hamstring muscles, and improves overall body balance and posture. **Q22.** An **Occupational Therapist** helps CWSN develop and improve their fine motor skills (like gripping, pinching, or hand-eye coordination) and aids them in performing daily life activities and handling sports equipment effectively. **Q23. Laceration** is a soft tissue injury characterized by an irregular, jagged tear in the skin caused by severe blunt trauma or a rough impact. **Example:** A football player sliding on a rough, stony ground, causing a deep, jagged tear on the knee. **Q24. Static Equilibrium** is when an athlete maintains stability while the body is completely at rest (e.g., a gymnast holding a handstand or a shooter aiming). **Dynamic Equilibrium** is maintaining balance while the body is in uniform motion (e.g., a cyclist riding a bike or a sprinter rounding a curve).

Q25. Matches Formula: $N(N-1)/2$ where N is the number of teams. **Rounds Formula (if N is even):** $N-1$. **Calculation for 8 teams:** Matches = $8(8-1)/2 = 8 \times 7/2 = 56/2 = 28$ matches. Rounds = $8-1=7$ rounds. **Q26.** The **Female Athlete Triad** is a severe health syndrome affecting female athletes, comprising three interconnected conditions driven by low energy availability. **Eating Disorders:** This is the primary trigger where an athlete drastically restricts caloric intake (anorexia) or purges food (bulimia) due to psychological pressure to maintain low body weight for aesthetic or weight-class sports, leading to severe nutritional deficiencies. **Q27.** 1. **"Carbohydrates make you fat":** Fact - Excess calories make you fat. Complex carbs are essential for sustained energy. 2. **"Starving/Skipping meals helps weight loss":** Fact - It lowers the Basal Metabolic Rate (BMR) and causes muscle loss, not healthy fat loss. 3. **"Fat-free products are always healthy":** Fact - Many fat-free processed foods compensate for lack of taste by adding excessive amounts of unhealthy refined sugars. **Q28. Procedure:** Participants run or walk the 600m distance on a track as fast as possible. The time taken to cover the distance is recorded in minutes and seconds. **Measurement:** This test specifically measures an athlete's Cardiovascular/Aerobic Endurance (the efficiency of the heart and lungs). **Q29.** 1. **Increased Heart Rate:** The heart beats much faster to supply oxygenated blood to working muscles. 2. **Increased Stroke Volume:** The amount of blood pumped per beat increases significantly. 3. **Increased Cardiac Output:** The total volume of blood pumped by the heart in one minute rises to meet the body's high oxygen demand. **Q30. Newton's First Law (Law of Inertia):** An object will remain at rest or in uniform motion in a straight line unless acted upon by an external resultant force. **Examples:** 1. A soccer ball placed on the penalty spot will remain completely stationary until a player kicks it. 2. A sprinter resting in the starting blocks will not move forward until they apply explosive muscular force against the blocks.

Q31. (i) Plate Tapping Test. (ii) Flamingo Balance Test. (iii) BMI = $\frac{\text{Weight (in kg)}}{\text{Height (in meters)}^2}$. (iv) It helps in the early identification of motor skill deficits or health issues like obesity, allowing for timely corrective measures and talent identification. **OR** The participant balances on one leg on a beam; the score is the number of falls/losses of balance recorded within 60 seconds. **Q32.** (i) Lactic Acid. (ii) Strain. (iii) Tidal Volume. (iv) Use the **PRICE** protocol: Protect the leg, Rest immediately, Apply Ice to reduce swelling, Compress with a bandage, and Elevate the leg. **OR** Cardiac Output. **Q33.** (i) Sports Equipment. (ii) Deaflympics (International Committee of Sports for the Deaf). (iii) Inclusive Education / Inclusion. (iv) Occupational Therapist. **OR** 8 years.

Q34. Calculations for 23 Teams (N=23): * Total Matches = $N-1=23-1=22$ matches. * Total Byes = Next power of 2 minus $N = 32-23 = 9$ Byes. * Teams in Upper Half = $(N+1)/2 = 24/2 = 12$ teams. * Teams in Lower Half = $(N-1)/2 = 22/2 = 11$ teams. * Byes in Upper Half = $(Nb-1)/2 = 8/2 = 4$ Byes. * Byes in Lower Half =

$(Nb + 1)/2 = 10/2 = 5$ Byes. * Total Rounds = $2^5 = 32$, so 5 Rounds. *Fixture Draw Setup:* The evaluator will check for a bracket of 23 teams. 12 teams in the top bracket, 11 in the bottom. 5 byes placed in the lower half (bottom, top of lower, etc.) and 4 byes in the upper half (top, bottom of upper, etc.). Remaining teams play Round 1. Winners advance up to Round 5 (Finals). **Q35. Diabetes** is a metabolic lifestyle disease characterized by high blood sugar levels resulting from the pancreas's inability to produce sufficient insulin. **Asanas to cure Diabetes:** Ardha Matsyendrasana, Bhujangasana, Paschimottanasana, Pavanmuktasana. **Procedure of Paschimottanasana:** Sit on the floor with both legs extended straight in front. Keep the spine erect. Inhale and raise both arms upward. Exhale and slowly bend forward from the hip joints. Try to hold the big toes with your fingers and touch your forehead to your knees without bending the knees. Hold the posture, breathe normally, then inhale and return. **Contraindications:** 1. Individuals suffering from severe slip disc or sciatica should avoid it. 2. Pregnant women or people with recent abdominal surgeries must not perform this pose. **Q36. Vitamins** are vital micro-nutrients required by the body in small quantities to protect against diseases, boost immunity, and act as catalysts for cellular metabolism. **Classification:** 1. **Fat-Soluble Vitamins:** Stored in the body's fatty tissues and liver. Includes: Vitamin A, Vitamin D, Vitamin E, Vitamin K. 2. **Water-Soluble Vitamins:** Cannot be stored in the body and dissolve in water. Includes: Vitamin B (Complex) and Vitamin C. **Deficiency Diseases:** * **Vitamin A:** Night blindness (inability to see in low light). * **Vitamin C:** Scurvy (bleeding gums, weak joints). * **Vitamin D:** Rickets (soft and deformed bones in children) or Osteomalacia (in adults). **Q37. Levers** are simple machines consisting of a rigid rod that rotates around a fixed axis (fulcrum) to overcome a resistance (load) using applied force (effort). **Class I Lever:** The Fulcrum is between the Effort and the Load. Affords balance. *Example:* Nodding the head backward and forward (Fulcrum is the neck joint, effort by neck muscles, load is the weight of the head). **Class II Lever:** The Load is located between the Fulcrum and the Effort. Affords mechanical force advantage. *Example:* Standing on tiptoes/Calf raises (Fulcrum is the toes, Load is body weight at the center of the foot, Effort is the gastrocnemius/calf muscle pulling the heel up). **Class III Lever:** The Effort is located between the Fulcrum and the Load. Affords speed and range of motion. *Example:* Bicep curl (Fulcrum is the elbow joint, Effort is the bicep muscle attached near the elbow, Load is the dumbbell in the hand).
